

J/ψ Discovery

1) e^+e^- collisions look σ for production

$$\left. \begin{aligned} \sigma(e^+e^- \rightarrow \text{had}) \\ \sigma(e^+e^- \rightarrow e^+e^-) \\ \sigma(e^+e^- \rightarrow \mu^+\mu^-) \end{aligned} \right\} \text{peak @ } \sqrt{s} = 3.016 \text{ GeV}$$

2) hadron collisions: $p + p \rightarrow X + \text{hadrons}$
 $\quad \quad \quad \hookrightarrow e^+e^-$

peak in M_{ee} @ 3.1 GeV

width of the peak dominated by experimental resolution.

$J/\psi : J^{PC} = 1^{--}$

sin σ_m dominates $\Rightarrow \Gamma_{J/\psi}$ has to be small

J/ψ width

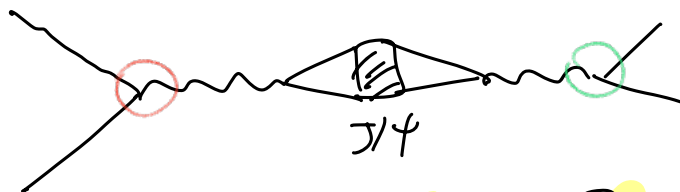
$$\Gamma_{\text{tot}} = \sum_i \Gamma_i$$

$\Gamma_i (J/\psi \rightarrow \text{final state } i)$

$$\text{BF}(J/\psi \rightarrow f_i) = \frac{\Gamma_i}{\Gamma_{\text{tot}}}$$

In e^+e^- experiment

$$\#(e^+e^- \rightarrow \text{had}) \propto \sigma(e^+e^- \rightarrow \text{had}) \propto \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{had}}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{tot}}^2}{(x-m)^2 + (\Gamma_{\text{tot}}/2)^2}$$



$$\#(e^+e^- \rightarrow \mu^+\mu^-) \propto \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\mu\mu}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{tot}}^2}{(x-m)^2 + (\Gamma_{\text{tot}}/2)^2}$$

$$\#(e^+e^- \rightarrow e^+e^-) \propto \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{ee}}{\Gamma_{\text{tot}}} \frac{\Gamma_{\text{tot}}}{(x-m)^2 + (\Gamma_{\text{tot}}/2)^2}$$

Experimental measurement of observables

$$\sigma(e^+e^- \rightarrow \text{had})$$

$$\sigma(e^+e^- \rightarrow \mu^+\mu^-)$$

$$\sigma(e^+e^- \rightarrow c^+\bar{c})$$

Parameters to obtain: 4 unknowns.

$$\Gamma_{\text{tot}}, \Gamma_{ee}, \Gamma_{\mu\mu}, \Gamma_{\text{had}}$$

1 assumption/constraint: $\Gamma_{\text{tot}} = \Gamma_{\text{had}} + \Gamma_{\mu\mu} + \Gamma_{ee}$

$\Rightarrow$ 4 equations w/ 4 unknowns.

$$\Rightarrow \Gamma_{\text{tot}} = 87 \text{ keV}$$

$$\frac{\Gamma}{M} = \frac{87 \text{ keV}}{3.1 \text{ GeV}} \quad \text{very narrow resonance}$$

e^+e^- machine $\sigma_E \sim 2.5 \text{ MeV}$

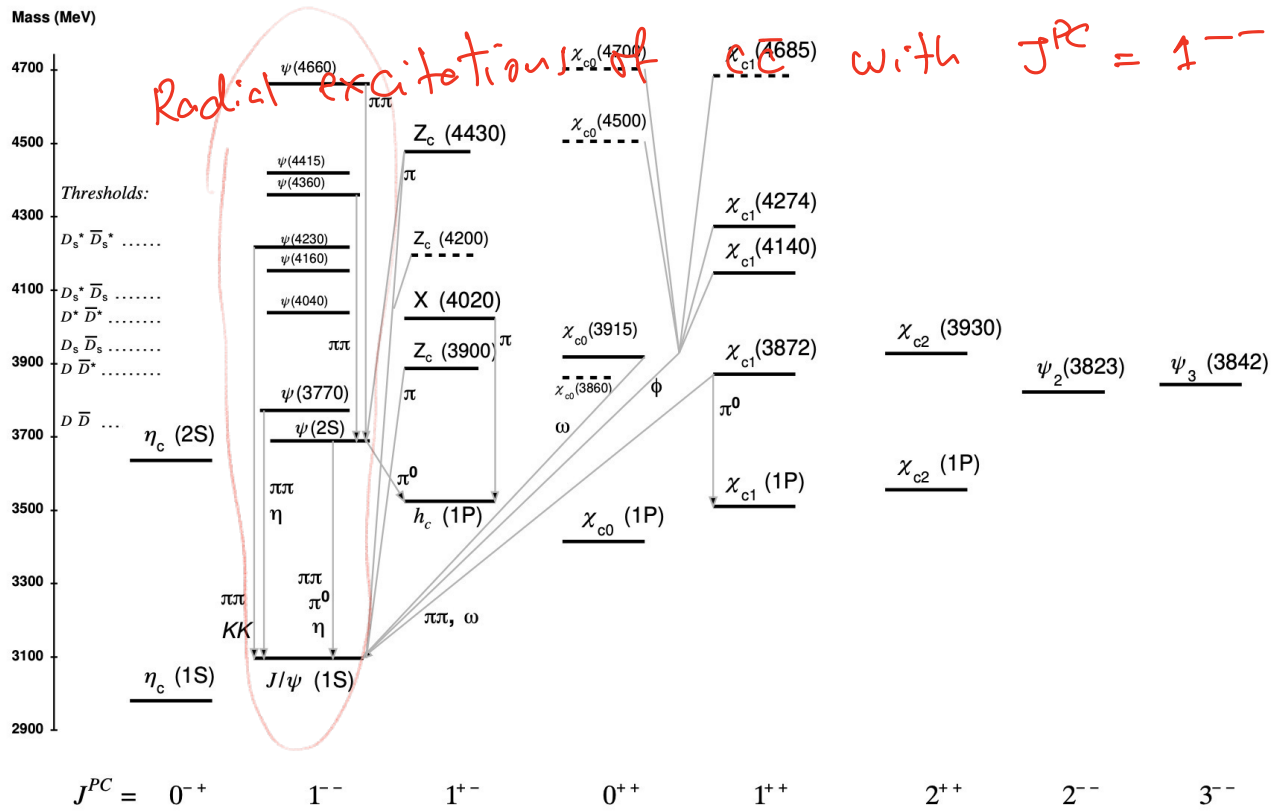
had machine $\sigma_E \sim 20 \text{ MeV}$

	mass	Γ	Flavor content	J^{PC}
$\rho(770)$	770 MeV	150 MeV	$(u\bar{u} - d\bar{d})/\sqrt{2}$	1^{--}
$\phi(1020)$	1020 MeV	4 MeV	$s\bar{s}$	1^{--}
J/ψ	3100 MeV	87 keV	$c\bar{c}$	1^{--}

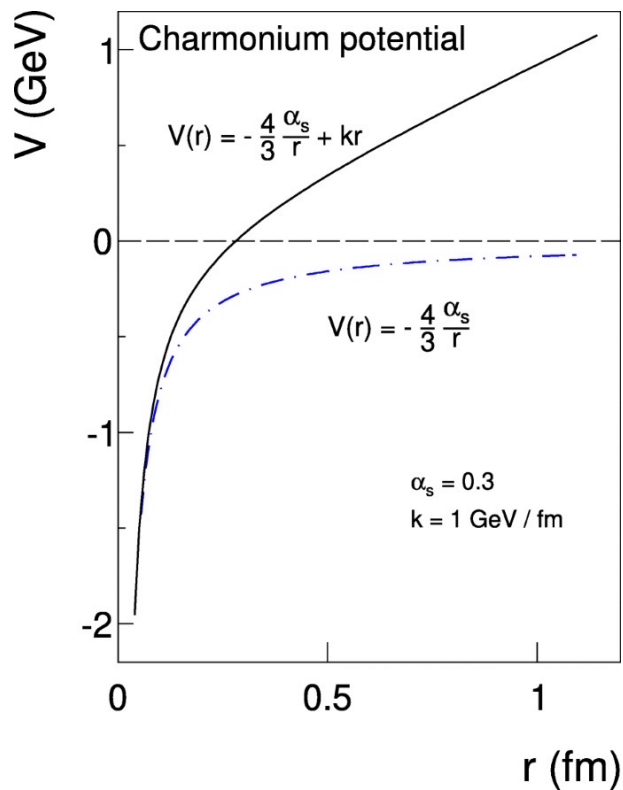
J/ψ is a very narrow resonance.

$\psi(2S)$ excited state of $c\bar{c}$ (charmonium)

if $c\bar{c}$ is a bound state $\Rightarrow$ are there other states?



Compute spectrum
of $c\bar{c}$ states
using QCD potential.



Why no $\eta_c \rightarrow \mu^+ \mu^-$
 $e^+ e^-$ } $Q > 0$

η_c $J^{PC} = 0^{-+}$

$P = (-1)^{L+1}$
 $S = 0$
 $C = (-1)^{L+S}$

$$\pi^0$$

$$S=0, L=0$$

$$J^{PC} = 0^{-+}$$

$$C_{\pi^0} = (C_\gamma)^2 = +1$$

$$m_{\pi^0} \approx 135 \text{ MeV}$$

$$M_{\mu} \approx 106 \text{ MeV}$$

$$\pi^0 \rightarrow \gamma \gamma$$

~~$$\mu^+ \mu^-$$~~

$$Q < 0$$

$$e^+ e^-$$

$$S_z^{\pi^0} = 0$$

$$e^+ \leftarrow \pi^0 \rightarrow e^-$$

In π^0 rest frame

$$S_z^{\text{fin}} = -1$$

$$E_{e\pm} = \frac{1}{2} m_{\pi} \approx 70 \text{ MeV}$$

$$\beta\gamma = \frac{p}{m} \approx \frac{70}{0.5} = 140$$

$$m_c: m \approx 2950 \text{ MeV}$$

$$e^+ \leftarrow \eta_c \rightarrow e^-$$

$$E_{e\pm} = \frac{1}{2} m_{\eta} \approx 1450 \text{ MeV}$$

$$\Rightarrow \beta\gamma \approx 3000$$

$$\eta_c(1S)$$

$$J^{PC} = 0^+(0^-+)$$

$\eta_c(1S)$ DECAY MODES

Mode	Fraction (Γ_i/Γ)	Scale factor/ Confidence level
Decays involving hadronic resonances		
Γ_1 $\eta'(958)\pi\pi$	(2.0 $\pm$ 0.4) %	S=1.4
Γ_2 $\eta'(958)K\bar{K}$	(1.73 $\pm$ 0.35) %	
Γ_3 $\eta'(958)\eta\eta$	(3.4 $\pm$ 0.6) $\times 10^{-3}$	
Γ_4 $\rho\rho$	(1.8 $\pm$ 0.4) %	
Γ_5 $K^{*}(892)^0 K^{-}\pi^{+}$ + c.c.	(1.8 $\pm$ 0.5) %	S=1.9
Γ_6 $K^{*}(892)\bar{K}^{*}(892)$	(7.0 $\pm$ 1.2) $\times 10^{-3}$	
Γ_7 $K^{*}(892)^0 \bar{K}^{*}(892)^0 \pi^{+}\pi^{-}$	(1.4 $\pm$ 0.6) %	
Decays into stable hadrons		
Γ_{37} $K\bar{K}\pi$	(7.1 $\pm$ 0.4) %	S=1.1
Γ_{38} $K\bar{K}\eta$	(1.32 $\pm$ 0.15) %	
Γ_{39} $\eta\pi^{+}\pi^{-}$	(1.6 $\pm$ 0.4) %	S=1.9
Γ_{40} $\eta 2(\pi^{+}\pi^{-})$	(4.3 $\pm$ 1.3) %	
Γ_{41} $K^{+}K^{-}\pi^{+}\pi^{-}$	(8.3 $\pm$ 1.8) $\times 10^{-3}$	
Γ_{42} $K^{+}K^{-}\pi^{+}\pi^{-}\pi^0$	(3.4 $\pm$ 0.6) %	
Γ_{43} $K^0 K^{-}\pi^{+}\pi^{-}\pi^{+}$ + c.c.	(5.4 $\pm$ 1.5) %	

$\eta_c \not\rightarrow e^+ e^-, \mu^+ \mu^-$ not observed before JCF although lighter

b quark and bottomonium ($b\bar{b}$)

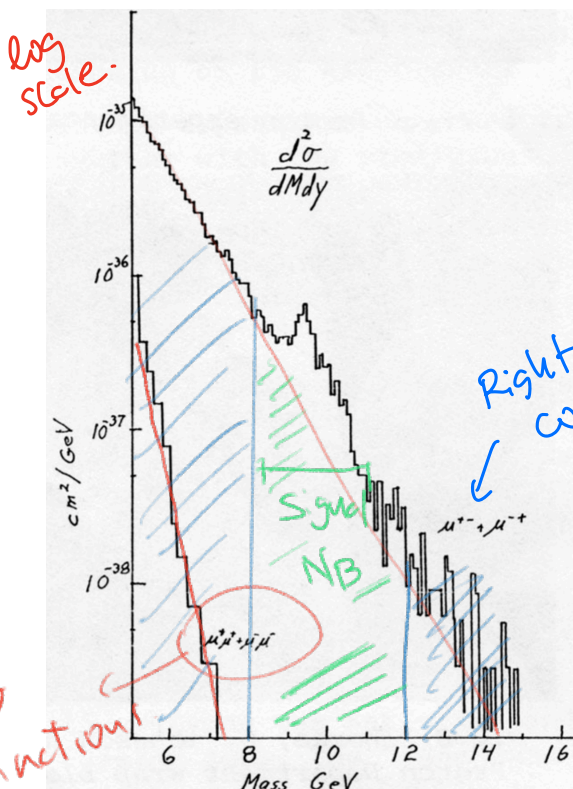
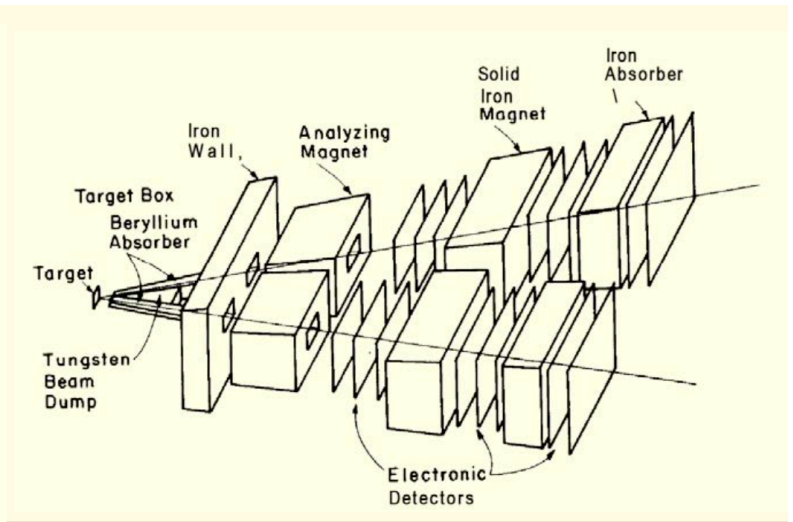
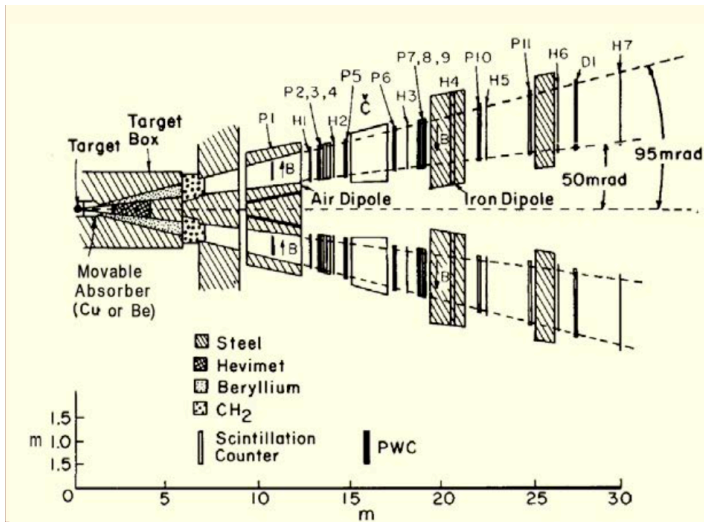
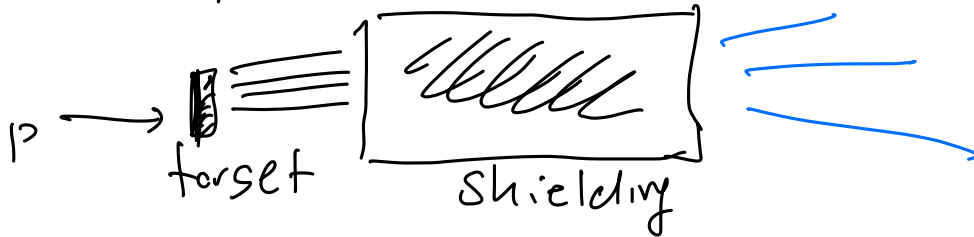
hadron production, Fermilab @ 1977

$$p + Cu(Pl) \rightarrow X + \text{hadrons}$$

$$\hookrightarrow \mu^+ \mu^-$$

look for peaks in invariant mass $m_{\mu\mu}$

$$p + Cu \rightarrow \pi^\pm + K^\pm + \pi^0 + p + \bar{p} + \text{had} + \mu^+ \mu^- \dots$$



$$p = 0.3 \frac{Z}{\text{GeV}} \frac{B}{T} \frac{R}{m}$$

No artificial peak in same-sign.

$\Rightarrow$ no mis-charge assignment

$$\gamma^{++} \rightarrow \mu^+ \mu^+$$

would give a peak.

no peak $\Rightarrow$ extract behaviour of background.

$$M_{\mu\mu} \sim 90 \text{ GeV} \Rightarrow P_{\mu} \sim \frac{1}{2} 10 \text{ GeV} = 5 \text{ GeV}.$$

$$P = 0.3 \times B \times R$$

$$R = \frac{S}{0.3 \times B} \sim \frac{10}{B}$$

$$\text{if } B = 1\% \Rightarrow R \sim 10\%$$

Background contribution under the peak from side bands.

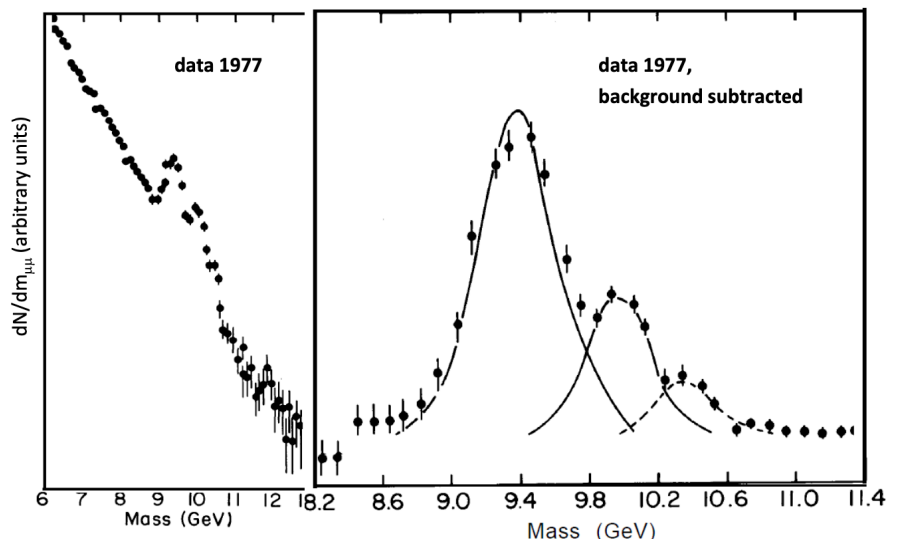
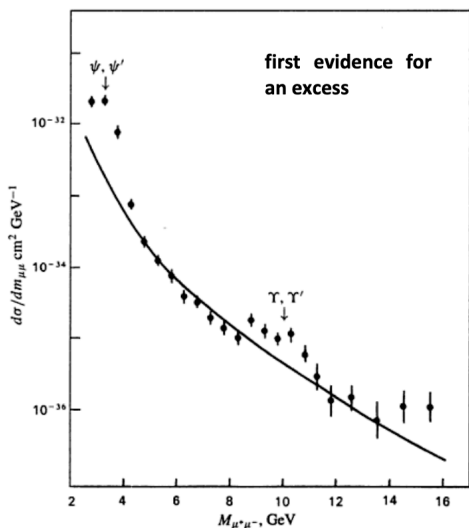
$$B(M_{\mu\mu}) \propto A e^{-\alpha M_{\mu\mu}}$$

Fit data in $[11, 16] + [5.5, 7.5]$ in $M_{\mu\mu}$

no excess in these side bands.

$\Rightarrow$ Fix/obtain α parameter for bkg

$$\Rightarrow N/B = \int_{\text{signal region}} dM_{\mu\mu} B(M_{\mu\mu})$$



Lederman
Nobel prize 1988

signal as sum
of multiple signals.

$$A e^{-\alpha m} + S_1 e^{-\frac{(m-m_1)^2}{2\sigma_1^2}} + S_2 e^{-\frac{(m-m_2)^2}{2\sigma_2^2}} + S_3 e^{-\frac{(m-m_3)^2}{2\sigma_3^2}}$$

α : nuisance param. to describe background

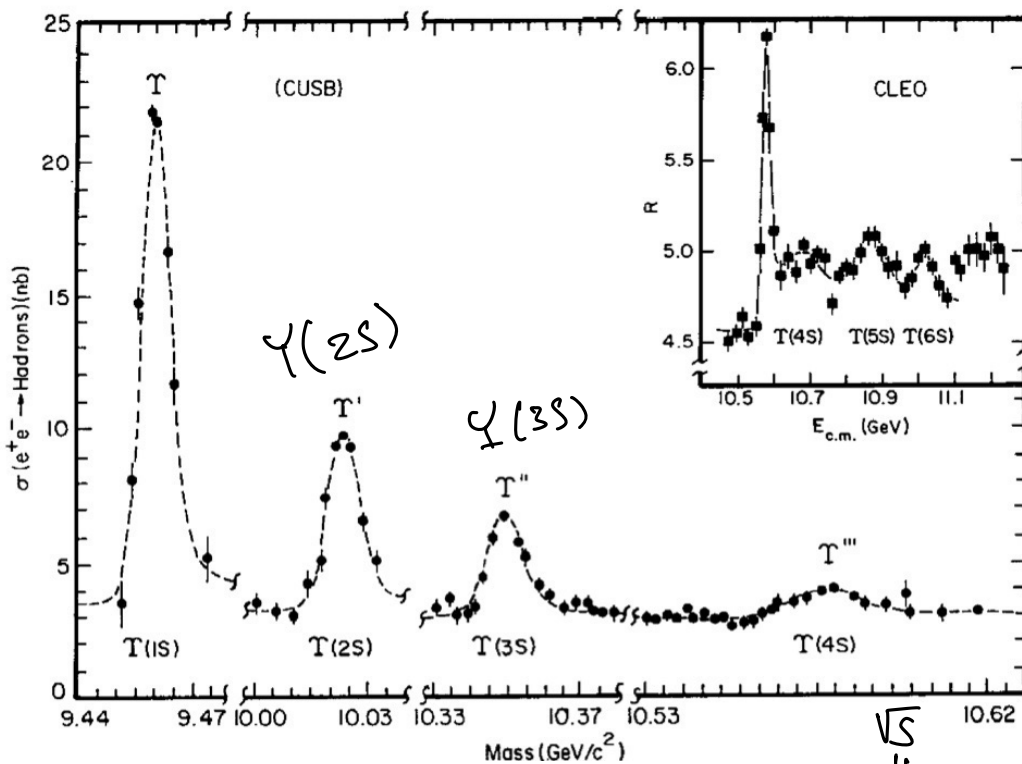
m_1, m_2, m_3 : mass of 3 states.

$\sigma_1, \sigma_2, \sigma_3$: dominated by exp. resolution

BW $\otimes$ Gaussian.

$\Rightarrow$ peaks due to $b\bar{b}$ state (Bottomonium)

study $b\bar{b}$ @ e^+e^- machine and scan σ vs $\sqrt{s}$

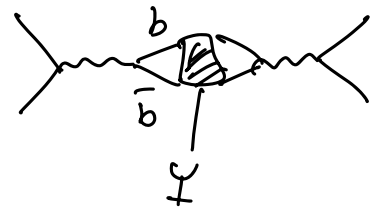


bottomonium

Upsilon

1S ground state

$$J^{PC} = 1^{--}$$



$$\psi(1S): \Gamma = 52 \text{ keV}$$

$$\psi(4S): \Gamma = 20.5 \text{ MeV}$$

$$\sqrt{s} = 2E_{beam}$$

$$m_X$$

$$\text{top quark: } e^+e^- \rightarrow t\bar{t} \rightarrow X$$

$$\sqrt{s} = 2E_{beam} > 2m_t \quad \sim 350 \text{ GeV}$$

$\Rightarrow E_{\text{beam}} = 180 \text{ GeV}$ not feasible with
small radius machines
because of beam synch.
rad.

$$P = 9 \text{ BR}$$

LHC: 27 km long not suff for e^+e^-

FCCee: Future Circular Collider $\sim 100 \text{ km}$.

LEP: e^+e^- collider 1989 $\rightarrow$ 2000

$$\sqrt{s} = 90 \text{ GeV} \rightarrow 202$$